

応用数学 3章 第9回 Basic 解答

【Basic】

問題 142

次の関数のフーリエ級数を求めよ. (1) $f(x) = \begin{cases} 2 & (-1 \leq x < 0) \\ 1 & (0 \leq x < 1) \end{cases}$, $f(x+2) = f(x)$ (2) $g(x) = \begin{cases} 3 & (-3 \leq x < 0) \\ x & (0 \leq x < 3) \end{cases}$, $g(x+6) = g(x)$ [教 p.82(問 3)]

解答:

$$\begin{aligned} (1) \quad c_0 &= \frac{1}{2} \int_{-1}^0 2 \, dx + \frac{1}{2} \int_0^1 1 \, dx = \frac{3}{2} \\ a_n &= \int_{-1}^0 2 \cos n\pi x \, dx + \int_0^1 \cos n\pi x \, dx = 0 \\ b_n &= \int_{-1}^0 2 \sin n\pi x \, dx + \int_0^1 \sin n\pi x \, dx = \\ &= \frac{2((-1)^n - 1)}{n\pi} + \frac{1 - (-1)^n}{n\pi} \\ &= \frac{n\pi}{(-1)^n - 1} \\ f(x) &= \frac{3}{2} - \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{\sin(2k+1)\pi x}{2k+1} \\ (2) \quad \text{周期 } 2l &= 6, \quad l = 3 \text{ として } c_0 = \frac{1}{6} \int_{-3}^0 3 \, dx + \\ &\frac{1}{6} \int_0^3 x \, dx = \frac{3}{2} + \frac{3}{4} = \frac{9}{4} \\ a_n &= \frac{1}{3} \int_{-3}^0 3 \cos \frac{n\pi x}{3} \, dx + \frac{1}{3} \int_0^3 x \cos \frac{n\pi x}{3} \, dx = \\ &\frac{3((-1)^n - 1)}{n^2 \pi^2} \\ b_n &= \frac{1}{3} \int_{-3}^0 3 \sin \frac{n\pi x}{3} \, dx + \frac{1}{3} \int_0^3 x \sin \frac{n\pi x}{3} \, dx = -\frac{3}{n\pi} \\ g(x) &= \frac{9}{4} - \frac{6}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos \frac{(2k+1)\pi x}{3} - \\ &\frac{3}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x}{3} \end{aligned}$$